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Process Concepts:

Program-1:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 processid.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("My Process ID (PID) = %d\n",getpid());
return 0;
}|
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc processid.c -o processid
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./processid
My Process ID (PID) = 620
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ |
```

Program-2:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 creating_child_using_fork.c
#include <stdio.h>
#include <unistd.h>
int main(){
pid_t pid=fork();
if(pid==0)
{
printf("Child Process\n");
printf("PID = %d\n", getpid());
}
else{
printf("Parent Process\n");
printf("PID = %d\n", getpid());
}
return 0;
}|
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ nano creating_child_using_fork.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc creating_child_using_fork.c -o creating_child_using_fork
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./creating_child_using_fork
Parent Process
PID = 737
Child Process
PID = 738
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ |
```

Program-3:

```
tejaswini@LAPTOP-S8US186U x + v
GNU nano 8.7.1 parent_waits_for_child.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    pid_t pid = fork();
    if(pid==0)
    {
        printf("Child Process Executing...\n");
        sleep(3);
        printf("Child Finished\n");
    }
    else{
        wait(NULL);
        printf("Parent Process Continues\n");
    }
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc parent_waits_for_child.c -o parent_waits_for_child
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./parent_waits_for_child
Child Process Executing...
Child Finished
Parent Process Continues
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ |
```

Program-4:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 multiple_child_process.c
#include <stdio.h>
#include <unistd.h>
int main()
{
for(int i=1;i<=3;i++){
if(fork()==0)
{
printf("Child %d PID = %d\n",i,getpid());
return 0;
}}
return 0;
}|
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc multiple_child_process.c -o multiple_child_process
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./multiple_child_process
Child 1 PID = 1209
Child 2 PID = 1210
Child 3 PID = 1211
```

Program-5:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 execute_linux_command_using_execvp.c *
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Before exec()\n");
execvp("ls","ls","-l",NULL);
printf("This will not execute.\n");
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ nano execute_linux_command_using_exclp.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc execute_linux_command_using_exclp.c -o execute_linux_command_using_exclp
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./execute_linux_command_using_exclp
Before exec()
total 100
-rwxr-xr-x 1 tejaswini tejaswini 16096 Aug 12 06:47 creating_child_using_fork
-rw-r--r-- 1 tejaswini tejaswini 223 Aug 12 06:46 creating_child_using_fork.c
-rwxr-xr-x 1 tejaswini tejaswini 16024 Aug 12 08:01 execute_linux_command_using_exclp
-rw-r--r-- 1 tejaswini tejaswini 155 Aug 12 08:01 execute_linux_command_using_exclp.c
-rwxr-xr-x 1 tejaswini tejaswini 16056 Aug 12 07:56 multiple_child_process
-rw-r--r-- 1 tejaswini tejaswini 158 Aug 12 06:55 multiple_child_process.c
-rwxr-xr-x 1 tejaswini tejaswini 16096 Aug 12 06:51 parent_waits_for_child
-rw-r--r-- 1 tejaswini tejaswini 257 Aug 12 06:50 parent_waits_for_child.c
-rwxr-xr-x 1 tejaswini tejaswini 16000 Aug 12 06:43 processid
-rw-r--r-- 1 tejaswini tejaswini 110 Aug 12 06:43 processid.c
```

Program-6:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 parent_and_child_executing_concurrently.c
#include <stdio.h>
#include <unistd.h>
int main()
{
fork();
for(int i=1;i<=5;i++)
{
printf("PID: %d Count: %d\n",getpid(),i);
sleep(1);
}
return 0;
}|
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc parent_and_child_executing_concurrently.c -o parent_and_child_executing_concurrently
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./parent_and_child_executing_concurrently
PID: 1395 Count: 1
PID: 1396 Count: 1
PID: 1395 Count: 2
PID: 1396 Count: 2
PID: 1395 Count: 3
PID: 1396 Count: 3
PID: 1395 Count: 4
PID: 1396 Count: 4
PID: 1396 Count: 5
PID: 1395 Count: 5
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ |
```

Program-7:

```
tejaswini@LAPTOP-S8US186U × + ▾
GNU nano 8.7.1 zombie_process.c
#include <stdio.h>
#include <unistd.h>
int main()
{
pid_t pid=fork();
if(pid==0)
{
printf("Child Exiting\n");
return 0;
}
else{
sleep(15);
printf("Parent Finished\n");
}
return 0;
}|
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ nano zombie_process.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc zombie_process.c -o zombie_process
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./zombie_process
Child Exiting
Parent Finished
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ |
```

Program-8:

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 orphan_process.c *
#include <stdio.h>
#include <unistd.h>
int main()
{
pid_t pid = fork();
if(pid>0)
{
printf("Parent Exiting\n");
return 0;
}
else
{
sleep(5);
printf("Child PID: %d\n",getpid());
printf("New Parent PID: %d\n", getppid());
}
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ nano orphan_process.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ gcc orphan_process.c -o
orphan_process
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ ./orphan_process
Parent Exiting
tejaswini@LAPTOP-S8US186U:~/OS_Lab/process_concepts$ Child PID: 1531
New Parent PID: 334
```